

Esame scritto MQ1 canale B - 16/06/2025

A scelta del candidato, si svolgano **tre esercizi** tra i quattro proposti. **Non** presentare un elaborato che contenga un numero maggiore di soluzioni rispetto alla richiesta: verranno considerate valide solo le prime tre risposte, nell'ordine in cui sono scritte nell'elaborato. La durata della prova scritta è di **3 ore**. Indicare su ogni foglio il proprio **nome, cognome e numero di matricola**.

Esercizio 1

Si consideri una buca di potenziale infinita in una dimensione, di larghezza a ($0 < x < a$), al cui interno è presente una particella di massa m . La particella si trova nello stato a energia minore, tale che:

$$E_1 = \frac{\hbar^2 \pi^2}{2ma^2}, \quad \psi_1(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi x}{a}\right). \quad (1)$$

Al tempo $t = 0$ la parete di destra della buca si sposta istantaneamente di modo che la larghezza della buca raddoppi ($0 < x < 2a$). Si assuma che questo spostamento *non* cambi lo stato iniziale della particella, che rimane lo stesso appena prima e appena dopo lo spostamento.

- a) Si determini la funzione d'onda del sistema al tempo $t > 0$ utilizzando le autofunzioni del nuovo hamiltoniano e si calcoli la probabilità P_n che la particella sia descritta da una arbitraria autofunzione del sistema modificato.
- b) Si calcoli il valore di aspettazione dell'hamiltoniano modificato al tempo $t > 0$.

Esercizio 2

L'hamiltoniano di un sistema a due livelli è descritto dalle seguenti relazioni:

$$H|1\rangle = |1\rangle + \frac{1+i}{\sqrt{2}}|2\rangle, \quad H|2\rangle = |2\rangle + \frac{1-i}{\sqrt{2}}|1\rangle, \quad (2)$$

dove $|1\rangle$ and $|2\rangle$ sono gli autostati normalizzati di un altro operatore hermitiano A :

$$A|1\rangle = \sqrt{2}|1\rangle, \quad A|2\rangle = -\sqrt{2}|2\rangle. \quad (3)$$

All'istante $t = 0$ si effettua una misura dell'osservabile associata ad A . Il valore ottenuto è $-\sqrt{2}$.

- a) Immediatamente dopo, si esegue una misura dell'energia del sistema. Qual è la probabilità di trovare il sistema nello stato fondamentale dell'hamiltoniano?
- b) Come cambierebbe questa probabilità se si eseguisse la misura dopo un intervallo di tempo T anziché immediatamente dopo?
- c) Se, invece, non si misurasse l'energia, in quali istanti di tempo il sistema si troverebbe nello stato $|2\rangle$?

Esercizio 3

Una particella di massa m soggetta al potenziale armonico

$$V(x) = \frac{1}{2} m \omega^2 x^2, \quad (4)$$

si trova, all'istante temporale $t = 0$, in uno stato determinato dalle seguenti condizioni:

- Ogni misura dell'energia fornisce valori E che rispettano la condizione

$$\hbar\omega < E < 3\hbar\omega \quad (5)$$

- Il valore di aspettazione dell'energia è

$$\langle E \rangle = \frac{11}{6} \hbar\omega \quad (6)$$

- Il valore di aspettazione della posizione è

$$\langle x \rangle = -\sqrt{\frac{8\hbar}{9m\omega}}. \quad (7)$$

Si determinino:

- lo stato in cui si trova il sistema a $t = 0$
- gli istanti di tempo in cui il valore di aspettazione della posizione è positivo e massimo.

Esercizio 4

Due particelle di massa m sono confinate a muoversi su una circonferenza di raggio R . L'operatore hamiltoniano che le governa è scritto in forma separabile:

$$H = H_1 + H_2, \quad H_i = -\frac{\hbar^2}{2mR^2} \frac{\partial^2}{\partial \phi_i^2} \quad (i = 1, 2), \quad (8)$$

dove ϕ_1 e ϕ_2 sono le coordinate angolari che identificano la posizione delle particelle sulla circonferenza. Essendo il moto su una circonferenza, ciascuna autofunzione di H_i soddisfa la condizione di periodicità:

$$\psi(\phi_i) = \psi(\phi_i + 2\pi) \quad (i = 1, 2). \quad (9)$$

- Si calcolino i livelli di energia e le autofunzioni di H .

Successivamente, viene inserita una interazione tra le due particelle descritta dal potenziale:

$$V = V_0 \cos(\phi_1 - \phi_2), \quad (10)$$

- Si introducano le variabili $\alpha = (\phi_1 + \phi_2)/2$ e $\beta = (\phi_1 - \phi_2)$. Si scriva l'equazione di Schrödinger indipendente dal tempo nelle variabili α e β , dimostrando che è separabile e si determini la periodicità della funzione d'onda in termini di queste variabili.
- Utilizzando le variabili α , β e assumendo che V sia una perturbazione, si calcoli la variazione al primo ordine per gli autovalori dell'energia.

I. FORMULE UTILI

Autofunzioni e autovalori per hamiltoniana di buca infinita 1D ($0 < x < a$):

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a}x\right), \quad (11)$$

$$E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}, \quad (12)$$

$$\sum_{\nu=0}^{+\infty} \frac{(2\nu+1)^2}{[(2\nu+1)^2-4]^2} = \frac{\pi^2}{16}, \quad (13)$$

$$\sum_{\nu=0}^{+\infty} \frac{1}{[(2\nu+1)^2-4]^2} = \frac{\pi^2}{64}. \quad (14)$$

$$(15)$$

Integrale gaussiano (per $a > 0$):

$$\int_{-\infty}^{+\infty} dx e^{-ax^2} = \sqrt{\frac{\pi}{a}} \quad (16)$$

Integrali del tipo:

$$\int_{-\infty}^{+\infty} dx e^{-(ax^2+bx)} \quad (17)$$

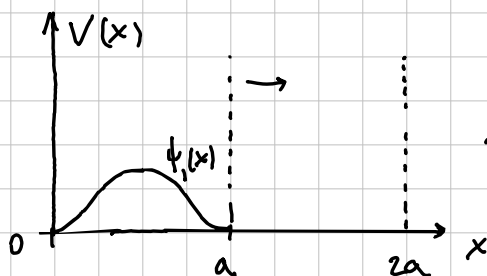
possono essere trattati completando il quadrato dell'argomento dell'esponenziale: prendere $y \equiv \sqrt{a}[x + (b/2a)]$ e notare che $(ax^2 + bx) = y^2 - b^2/4a$.

QUANTUM MECHANICS

1-B

Written exam, 16/06/2025

{ EXERCISE 1 }



{ Griffith's form:
 $0 < x < a$ }

$$E_1 = \frac{\hbar^2 \pi^2}{2ma^2}, \quad \psi_1(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{\pi}{a} x\right)$$

MODIFIED SYSTEM : $a \rightarrow 2a$

$$E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2} \longrightarrow \tilde{E}_n = \frac{n^2 \hbar^2 \pi^2}{8ma^2} = \frac{1}{4} E_n$$

$$\begin{aligned} \psi_n &= \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi}{a} x\right) \longrightarrow \tilde{\psi}_n = \frac{1}{\sqrt{a}} \sin\left(\frac{n\pi}{2a} x\right) = \\ &= \frac{1}{\sqrt{2}} \psi_n(x/2) \end{aligned}$$

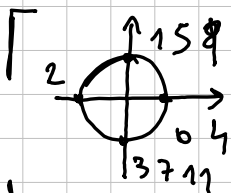
$$\tilde{\psi}(x, t=0) = \psi_n(x)$$

$$a) \tilde{\psi}(x, t) = \sum_{n=1}^{+\infty} c_n e^{-i \frac{\tilde{E}_n}{\hbar} t} \tilde{\psi}_n(x)$$

$$c_n = \langle \tilde{\psi}_n | \tilde{\psi}(x, 0) \rangle = \int_0^{2a} dx \tilde{\psi}_n^*(x) \tilde{\psi}(x, 0) = \longrightarrow$$

$$\begin{aligned}
 &= \int_0^a dx \tilde{\psi}_n^*(x) \psi_1(x) = \\
 &= \int_0^a dx \frac{1}{\sqrt{2}} \psi_n^*(x/2) \psi_1(x) = \\
 &= \frac{1}{\sqrt{2}} \frac{2}{a} \int_0^a dx \sin\left(\frac{\pi n}{2a} x\right) \sin\left(\frac{\pi}{a} x\right) =
 \end{aligned}$$

$$\begin{aligned}
 \sin \alpha \cdot \sin \beta &= \frac{1}{2} \left[\cos(\alpha - \beta) - \cos(\alpha + \beta) \right] \\
 &= \frac{\sqrt{2}}{a} \frac{1}{2} \int_0^a dx \left[\cos\left(\frac{\pi n x - 2\pi x}{2a}\right) - \cos\left(\frac{\pi n x + 2\pi x}{2a}\right) \right] = \\
 &\quad x' = \frac{\pi x}{2a} (n-2) \quad x' = \frac{\pi x}{2a} (n+2) \\
 &= \frac{\sqrt{2}}{2a} \left[\frac{2a}{\pi(n-2)} \int_0^{\pi(n-2)/2} dx' \cos x' - \frac{2a}{\pi(n+2)} \int_0^{\pi(n+2)/2} dx' \cos x' \right] = \\
 &= \frac{\sqrt{2}}{\pi(n-2)} \sin\left[\frac{\pi}{2}(n-2)\right] - \frac{\sqrt{2}}{\pi(n+2)} \sin\left[\frac{\pi}{2}(n+2)\right].
 \end{aligned}$$



$$\sin \frac{\pi}{2} k = (-1)^{(k-1)/2}, \quad k > 1 \text{ and } \underline{\text{odd}}$$

$$c_n = \frac{\sqrt{2}}{\pi} \left\{ \frac{1}{(n-2)} \sin\left(\frac{\pi}{2}(n-2)\right) - \frac{1}{(n+2)} \sin\left(\frac{\pi}{2}(n+2)\right) \right\}$$

For odd n :

$$c_n^{\text{odd}} = \frac{\sqrt{2}}{\pi} \left\{ \frac{1}{n-2} (-)^{(n-3)/2} - \frac{1}{n+2} (-)^{(n+1)/2} \right\}$$

$$P_n^{\text{odd}} = |c_n|^2 = \frac{2}{\pi^2} \left[\frac{(-)^{n-3}}{(n-2)^2} + \frac{(-)^{n+1}}{(n+2)^2} + \frac{2}{(n-2)(n+2)} (-)^{(n-3)/2} \cdot (-)^{(n+1)/2} \right] =$$

$$\downarrow \frac{n}{2} - \frac{3}{2} \quad \frac{n}{2} + \frac{1}{2} \quad n-1$$

$$(-)^{\frac{n}{2} - \frac{3}{2}} (-)^{\frac{n}{2} + \frac{1}{2}} = (-)^{n-1}$$

$$= \frac{2}{\pi^2} \frac{1}{(n+2)^2(n-2)^2} \left\{ (n+2)^2 + (n-2)^2 - 2 \overbrace{(n+2)(n-2)}^{n^2-4} \right\} =$$

$$\cancel{n^2+4} + \cancel{4n^2+4} - \cancel{4n} - 2n^2 + 8$$

$$= \frac{32}{\pi^2 (n^2-4)^2}$$

$\frac{\pi^2}{64}$ (formula util.)

$$\sum_{n \geq 1, \text{ odd}} P_n^{\text{odd}} = \frac{32}{\pi^2} \sum_{l=0}^{\infty} \frac{1}{[(2l+1)^2-4]^2} = \frac{32}{\pi^2} \cdot \frac{\pi^2}{64} = \frac{1}{2}$$

→ let's now consider c_n , with n even

$$c_n = \frac{\sqrt{2}}{\pi} \left\{ \frac{1}{(n-2)} \sin\left(\frac{\pi}{2}(n-2)\right) - \frac{1}{(n+2)} \sin\left(\frac{\pi}{2}(n+2)\right) \right\}$$

$$c_2 = \frac{\sqrt{2}}{\pi} \left\{ \frac{1}{n-2} \sin\left(\frac{\pi}{2}(n-2)\right) - 0 \right\} \stackrel{\downarrow}{=} \sin x = x + o(x)$$

$$= \frac{\sqrt{2}}{\cancel{\pi}} \frac{1}{\cancel{n-2}} \cdot \frac{\cancel{\pi}}{2} \cancel{(n-2)} = \frac{1}{\sqrt{2}}$$

$$c_n^{\text{even}} = 0 \quad \forall n$$

$$p_n^{\text{even}} = |c_2|^2 = \frac{1}{2}$$

$$\Rightarrow \sum_{n=1}^{\infty} (p_n^{\text{odd}} + p_n^{\text{even}}) = \frac{1}{2} + \frac{1}{2} = 1 \quad \checkmark \quad \underline{\text{ok}}$$

$$\begin{aligned}
 b) \quad \langle \tilde{H} \rangle &= \langle \tilde{\psi}(x,t) | \tilde{H} | \tilde{\psi}(x,t) \rangle = \\
 &= \int_0^{2a} dx \left(\sum_n c_n^* e^{i \frac{\tilde{E}_n}{\hbar} t} \tilde{\psi}_n^*(x) \right) \left(\sum_m e^{-i \frac{\tilde{E}_m}{\hbar} t} \tilde{H} \tilde{\psi}_m(x) \right) = \\
 &= \sum_{n,m} c_n^* c_m \tilde{E}_m e^{i \frac{1}{\hbar} (\tilde{E}_n - \tilde{E}_m) t} \underbrace{\int_0^{2a} dx \tilde{\psi}_n^*(x) \tilde{\psi}_m(x)}_{\delta_{nm}} =
 \end{aligned}$$

$$= \sum_{n=1}^{+\infty} |c_n|^2 \tilde{E}_n.$$

$$\rightarrow \sum_{\substack{n=1 \\ (\text{odd})}}^{+\infty} \frac{\cancel{3^2}}{\pi^2 (n^2 - 4)^2} \frac{n^2 \cancel{4^2} \pi^2}{8 m a^2} = \frac{4 \hbar^2}{m a^2} \left[\sum_{n'=1}^{+\infty} \frac{(2n'+1)^2}{[(2n'+1)^2 - 4]^2} \right] =$$

$$= \frac{4 \hbar^2}{m a^2} \frac{\pi^2}{16} = \frac{\pi^2 \hbar^2}{4 m a^2} = \frac{1}{2} E_1$$

$$\begin{aligned}
 \rightarrow \sum_{\substack{n=1 \\ (\text{even})}}^{+\infty} |c_n|^2 \tilde{E}_n &= |c_2|^2 \tilde{E}_2 = \frac{1}{2} \frac{1}{4} E_1 = \frac{1}{2} \frac{1}{4} 4 E_1 \\
 &= \frac{1}{2} E_1
 \end{aligned}$$

$$\Rightarrow \langle \tilde{H} \rangle = \sum_{n=1}^{+\infty} |c_n|^2 \tilde{E}_n = \frac{1}{2} E_1 + \frac{1}{2} E_1 = E_1$$

↳ The energy of the initial state ($t=0$) is conserved;
ok, since $\frac{\partial}{\partial t} H = 0$.

{ EXERCISE 2 } ~ 2-LEVEL SYSTEM -

a) $\{|1\rangle, |2\rangle\}$ basis for the space (eigenvectors of an hermitean operator, A)

$$\langle 1|H|1\rangle = 1, \quad \langle 1|H|2\rangle = \frac{1-i}{\sqrt{2}}$$

$$\langle 2|H|1\rangle = \frac{1+i}{\sqrt{2}}, \quad \langle 2|H|2\rangle = 1$$

In the chosen basis:

$$H = \begin{pmatrix} 1 & \frac{1-i}{\sqrt{2}} \\ \frac{1+i}{\sqrt{2}} & 1 \end{pmatrix}$$

$$0 = \det(H - \lambda \mathbb{1}) = \det \begin{pmatrix} 1-\lambda & \frac{1-i}{\sqrt{2}} \\ \frac{1+i}{\sqrt{2}} & 1-\lambda \end{pmatrix} =$$

$$= (1-\lambda)^2 - \frac{1}{2} (1-i)(1+i) = 1 + \lambda^2 - 2\lambda - 1 =$$

$$= \lambda(\lambda - 2)$$

→ eigenvalues: $\lambda_- = 0$, $\lambda_+ = 2$ (energy levels)

→ ground state

$$|\psi_-\rangle, \lambda_- = 0:$$

$$H|\psi_-\rangle = \lambda_-|\psi_-\rangle = 0$$

$$\begin{pmatrix} 1 & \frac{(1-i)}{\sqrt{2}} \\ \frac{1+i}{\sqrt{2}} & 1 \end{pmatrix} \begin{pmatrix} \psi_{-1} \\ \psi_{-2} \end{pmatrix} = 0$$

$$\begin{cases} \psi_{-1} + \frac{1}{\sqrt{2}}(1-i)\psi_{-2} = 0 \\ \frac{1+i}{\sqrt{2}}\psi_{-1} + \psi_{-2} = 0 \end{cases} \rightarrow \psi_{-1} = \alpha, \text{ constant}$$

$$|\psi_-\rangle = \alpha \begin{pmatrix} 1 \\ -\frac{1+i}{\sqrt{2}} \end{pmatrix} \quad \propto \begin{pmatrix} -\frac{1-i}{\sqrt{2}} \\ 1 \end{pmatrix}$$

$$1 = \langle \psi_- | \psi_- \rangle = |\alpha|^2 \left(1 + \frac{1}{2}(1+i)(1-i) \right) = 2|\alpha|^2$$

$\alpha = \frac{1}{\sqrt{2}} \rightarrow$ a global phase would be irrelevant

$$|\psi_-\rangle = \begin{pmatrix} 1/\sqrt{2} \\ -(1+i)/2 \end{pmatrix}, \text{ ground state}$$

→ excited state $|\psi_+\rangle$, $\lambda_+ = 2$:

$$H|\psi_+\rangle = \lambda_+ |\psi_+\rangle$$

$$\begin{pmatrix} 1 & (1-i)/\sqrt{2} \\ (1+i)/\sqrt{2} & 1 \end{pmatrix} \begin{pmatrix} \psi_{+1} \\ \psi_{+2} \end{pmatrix} = 2 \begin{pmatrix} \psi_{+1} \\ \psi_{+2} \end{pmatrix}$$

$$\begin{cases} \psi_{+1} + \frac{1}{\sqrt{2}}(1-i)\psi_{+2} = 2\psi_{+1} \\ (1+i)\frac{\psi_{+1}}{\sqrt{2}} + \psi_{+2} = 2\psi_{+2} \end{cases} \rightarrow \begin{aligned} \psi_{+1} &= \alpha, \text{ constant} \\ \psi_{+2} &= \frac{1+i}{\sqrt{2}}\alpha \end{aligned}$$

$$|\psi_+\rangle = \alpha \begin{pmatrix} 1 \\ \frac{1+i}{\sqrt{2}} \end{pmatrix}$$

$$1 = \langle \psi_+ | \psi_+ \rangle = |\alpha|^2 \left(1 + \frac{1}{2}(1+i)(1-i) \right) = 2|\alpha|^2$$

$$\alpha = 1/\sqrt{2} \rightarrow \text{global phase irrelevant}$$

$$|\psi_+\rangle = \begin{pmatrix} 1/\sqrt{2} \\ (1+i)/2 \end{pmatrix}, \text{ excited state}$$

Thus:

$$|\psi_-\rangle = \begin{pmatrix} 1/\sqrt{2} \\ -(1+i)/2 \end{pmatrix} \equiv \frac{1}{\sqrt{2}} |1\rangle - \frac{1+i}{2} |2\rangle$$

$$|\psi_+\rangle = \begin{pmatrix} 1/\sqrt{2} \\ (1+i)/2 \end{pmatrix} = \frac{1}{\sqrt{2}} |1\rangle + \frac{(1+i)}{2} |2\rangle$$

→ Measure A and find $-\sqrt{2} \Rightarrow |\psi\rangle = |2\rangle$

→ Measure H; probability of getting $\{\lambda_-, |\psi_-\rangle\}$:

$$\begin{aligned} P(E = \lambda_-) &= |\langle \psi_- | \psi \rangle|^2 = |\langle \psi_- | 2 \rangle|^2 = \\ &= \left| \frac{1}{\sqrt{2}} \langle 1 | 2 \rangle - \frac{1-i}{2} \underbrace{\langle 2 | 2 \rangle}_1 \right|^2 = \left| \frac{1-i}{2} \right|^2 = \\ &= \frac{2}{4} = \frac{1}{2} \end{aligned}$$

b) Measuring H after T rather than immediately after A:

$$|\psi(T)\rangle = e^{-i \frac{\hat{H}}{\hbar} T} |2\rangle = ?$$

Let's write $|2\rangle$ as a function of the eigenstates of H, for which we can easily calculate the time evolution:

$$|\psi_+\rangle + |\psi_-\rangle = \sqrt{2} |1\rangle$$

$$|\psi_+\rangle - |\psi_-\rangle = (1+i) |2\rangle$$



$$|2\rangle = \frac{1}{1+i} \{ |\psi_+\rangle - |\psi_-\rangle \} =$$

$$= \frac{1-i}{2} \{ |\psi_+\rangle - |\psi_-\rangle \}$$

$$|\psi(T)\rangle = e^{-i\frac{\hat{H}}{\hbar}T} |2\rangle =$$

$$= \frac{1-i}{2} \left\{ e^{-i\lambda_+T/\hbar} |\psi_+\rangle - \frac{e^{-i\lambda_-T/\hbar}}{1} |\psi_-\rangle \right\} =$$

$$= \frac{1-i}{2} \left\{ e^{-2iT/\hbar} |\psi_+\rangle - |\psi_-\rangle \right\}$$

$$P(E = \lambda_-, t = T) = |\langle \psi_- | \psi(T) \rangle|^2 = \frac{|1-i|^2}{4} = \frac{1}{2}$$

→ the probability doesn't change.

c) Find t such that $|\psi(t)\rangle \equiv |2\rangle$ (no measurements)

$$|\psi(t)\rangle = \frac{1-i}{2} \left\{ e^{-2it/\hbar} |\psi_+\rangle - |\psi_-\rangle \right\}$$

$$|2\rangle = \frac{1-i}{2} \left\{ |\psi_+\rangle - |\psi_-\rangle \right\}$$

$$(n=0 \rightarrow t=0)$$

$$h=1, 2, 3, \dots$$

$$\boxed{t = \pi \hbar n}$$

$$|\psi(t)\rangle \equiv |2\rangle \Leftrightarrow e^{-2it/\hbar} = 1 \rightarrow \frac{2t}{\hbar} = 2\pi n$$

{ EXERCISE 3 }

particle with mass m in harmonic 1-D potential :

$$V(x) = \frac{1}{2} m \omega^2 x^2$$

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega, \quad \hat{H} |n\rangle = E_n |n\rangle$$

$$\boxed{t=0} \rightarrow$$

$$1) \quad \hbar \omega < E < 3 \hbar \omega$$

$$2) \quad \langle E \rangle = \frac{11}{6} \hbar \omega$$

$$3) \quad \langle x \rangle = - \left(8 \hbar / 9 m \omega\right)^{1/2}$$

a) find the state of the system at $t=0$

$$1) \Rightarrow n=1, 2 \quad (\text{allowed values})$$

$$|\psi\rangle = c_1 |1\rangle + c_2 |2\rangle,$$

$$\text{with } |c_1|^2 + |c_2|^2 = 1 \quad (\text{normalization})$$

$$2) \quad \langle E \rangle = |c_1|^2 E_1 + |c_2|^2 E_2$$

$$\langle E \rangle = \frac{11}{6} \hbar \omega$$

$$\begin{cases} \frac{11}{6} \hbar \omega = |c_1|^2 \frac{3}{2} \hbar \omega + |c_2|^2 \frac{5}{2} \hbar \omega \\ 1 = |c_1|^2 + |c_2|^2 \end{cases}$$

$$\frac{11}{6} \cancel{\hbar \omega} = \frac{1}{2} \cancel{\hbar \omega} \left[3|c_1|^2 + 5(1 - |c_1|^2) \right]$$

$$\frac{11}{6} = \frac{1}{2} (-2|c_1|^2 + 5)$$

$$\frac{11}{3} - 5 = -2|c_1|^2, \quad \frac{4}{3} = 2|c_1|^2$$

$$\frac{2}{3} = |c_1|^2 \Rightarrow |c_2|^2 = \frac{1}{3}$$

$$\Rightarrow c_1 = \sqrt{\frac{2}{3}}, \quad c_2 = \frac{1}{\sqrt{3}} e^{i\alpha}$$

α is the relative phase between the states $|1\rangle, |2\rangle$; we fix it using condition 3):

$$3) \quad \hat{x} = \left(\frac{\hbar}{2m\omega} \right)^{1/2} (\hat{a} + \hat{a}^\dagger)$$

$$\begin{aligned} \langle x \rangle &= \langle \psi | \hat{x} | \psi \rangle = \\ &= - (8\hbar / 9m\omega)^{1/2} \end{aligned}$$

$$\langle x \rangle = (c_1^* \langle 1| + c_2^* \langle 2|) \sqrt{\frac{\hbar}{2m\omega}} \cdot (\hat{a} + \hat{a}^\dagger) \cdot$$

$$\cdot (c_1 |1\rangle + c_2 |2\rangle) =$$

$$= \sqrt{\frac{\hbar}{2m\omega}} (c_1^* \langle 1| + c_2^* \langle 2|) \left(0 + c_2 \sqrt{2} |1\rangle + c_1 \sqrt{2} |2\rangle + c_2 \sqrt{3} |3\rangle \right) =$$

$$\begin{cases} \hat{a} |n\rangle = \sqrt{n} |n-1\rangle \\ \hat{a}^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle \end{cases}$$

$$= \sqrt{\frac{\hbar}{2m\omega}} (c_1^* c_2 \sqrt{2} + c_2^* c_1 \sqrt{2}) =$$

$$= \sqrt{\frac{\hbar}{m\omega}} (c_1^* c_2 + c_1 c_2^*) \equiv \frac{-2}{3} \sqrt{\frac{2\hbar}{m\omega}}$$

$$\Rightarrow c_1^* c_2 + c_1 c_2^* \equiv \frac{-2\sqrt{2}}{3}$$

$$\frac{\sqrt{2}}{3} (e^{+i\alpha} + e^{-i\alpha}) = -2 \frac{\sqrt{2}}{3} \quad \cos \alpha = -1$$

$$\Rightarrow \alpha = \pi, \quad \left[c_1 = \sqrt{\frac{2}{3}}, \quad c_2 = -1/\sqrt{3} \right]$$

$$|4\rangle = \sqrt{\frac{2}{3}} |1\rangle - \frac{1}{\sqrt{3}} |2\rangle$$

b) Find the t values in which $\langle x \rangle(t)$ is positive and maximum

$$\begin{aligned} |4\rangle(t) &= e^{-\frac{i}{\hbar} \hat{H} t} |4\rangle = \\ &= \underbrace{\sqrt{\frac{2}{3}} e^{-\frac{i}{\hbar} E_1 t}}_{c_1(t)} |1\rangle - \underbrace{\frac{1}{\sqrt{3}} e^{-\frac{i}{\hbar} E_2 t}}_{c_2(t)} |2\rangle, \end{aligned}$$

$$E_1 = \frac{3}{2} \hbar \omega, \quad E_2 = \frac{5}{2} \hbar \omega$$

$$\begin{aligned} \langle x \rangle(t) &= \sqrt{\frac{\hbar}{m\omega}} \left[c_1^*(t) c_2(t) + c_1(t) c_2^*(t) \right] = \\ &= \sqrt{\frac{\hbar}{m\omega}} \left[-\frac{\sqrt{2}}{3} \left(e^{i\frac{3}{2}\omega t} e^{-i\frac{5}{2}\omega t} \right) - \frac{\sqrt{2}}{3} \left(e^{-i\frac{3}{2}\omega t} e^{i\frac{5}{2}\omega t} \right) \right] = \\ &= \sqrt{\frac{\hbar}{m\omega}} \left(-\frac{\sqrt{2}}{3} \right) \left(e^{-i\omega t} + e^{i\omega t} \right) = \\ &= -\frac{2}{3} \sqrt{\frac{2\hbar}{m\omega}} \cos \omega t = -k \cos \omega t, \quad k > 0 \end{aligned}$$

$$\langle x \rangle(t) = -k \cos \omega t, \quad k > 0$$

$\langle x \rangle(t)$ is positive and maximum when

$$\omega t = (2n+1)\pi, \quad n=0, 1, 2, \dots$$

$$t = \frac{(2n+1)\pi}{\omega} \quad //$$

— 0 —

{ EXERCISE 4 }

1-D perturbation
on a circle

R : radius of the circle

$$\hat{H} = \hat{H}_1 + \hat{H}_2, \quad \hat{H}_i = -\frac{\hbar^2}{2mR^2} \frac{\partial^2}{\partial \phi_i^2}, \quad i: 1, 2$$

$\phi_{1,2}$: angular positions on the circle

$\hat{H}_i$ -eigenfunctions are such that

$$\psi(\phi_i) \equiv \psi(\phi_i + 2\pi), \quad i: 1, 2$$

a) since the hamiltonian is written as a sum $\hat{H} = \hat{H}_1 + \hat{H}_2$, the eigenfunction of $\hat{H}$ will be a product of the single-particle wavefunctions:

$$\Psi_{k,l}(\phi_1, \phi_2) = \psi_k(\phi_1) \psi_l(\phi_2)$$

$$E_{k,l} = E_k + E_l$$

where $\psi_k(\phi_1) = ?$, $E_k = ?$ 

$$-\frac{\hbar^2}{2mR^2} \frac{d^2}{d\phi_1^2} \psi(\phi_1) = E \psi(\phi_1)$$

$$\psi''(\phi_1) = -\frac{2mR^2}{\hbar^2} E \psi(\phi_1), \quad \psi = e^{i\alpha\phi_1}$$

$$-\alpha^2 e^{i\alpha\phi_1} = -\frac{2mR^2}{\hbar^2} E e^{i\alpha\phi_1}$$

$$\alpha^2 = \frac{2mR^2}{\hbar^2} E \rightarrow \psi(\phi_1) \sim e^{\pm i\alpha\phi_1}$$

$$\psi(\phi_1) = \psi(\phi_1 + 2\pi) \rightarrow e^{i\alpha\phi_1} = e^{i\alpha\phi_1} \underbrace{e^{i\alpha 2\pi}}_1$$

$$\alpha = 0, \pm 1, \pm 2, \dots$$

α : let's call it $n \in \mathbb{Z}$

$$\left. \begin{aligned} E_n &= \frac{\hbar^2 n^2}{2mR^2} \\ \psi_n(\phi_1) &= c_n e^{in\phi_1} \end{aligned} \right\} n \in \mathbb{Z} \quad (n=0, \pm 1, \pm 2, \dots)$$

$$\rightarrow c_n = \frac{1}{\sqrt{2\pi}} \quad (\text{norm.})$$

$$\left\{ \begin{aligned} \Psi(\phi_1, \phi_2) &= \psi_n(\phi_1) \psi_\ell(\phi_2) \\ E_{n,\ell} &= \frac{\hbar^2}{2mR^2} (n^2 + \ell^2) \end{aligned} \right.$$

b) now let's introduce the interaction:

$$V(\phi_1, \phi_2) = V_0 \cos(\phi_1 - \phi_2) ,$$

and the hamiltonian is NOT separable in ϕ_1, ϕ_2 anymore.

Is there a set of variables in which the new hamiltonian is separable?

$$V = V(\beta) , \quad \beta \equiv \phi_1 - \phi_2$$

So let's use $\beta \equiv \phi_1 - \phi_2$ and find the other one, α , looking at the UNPERTURBED H :

$$\alpha \equiv \frac{1}{2}(\phi_1 + \phi_2) , \quad \beta = \phi_1 - \phi_2$$

$$H = - \frac{\hbar^2}{2mR^2} \left(\frac{\partial^2}{\partial \phi_1^2} + \frac{\partial^2}{\partial \phi_2^2} \right)$$

Let's write the derivatives as a function of α, β :

$$\phi_1 = \phi_1(\alpha, \beta)$$

$$\phi_2 = \phi_2(\alpha, \beta)$$

$$\alpha = \frac{1}{2}(\phi_1 + \phi_2), \quad \beta = \phi_1 - \phi_2$$

$$\frac{\partial}{\partial \phi_1} = \frac{\partial}{\partial \alpha} \frac{\partial \alpha}{\partial \phi_1} + \frac{\partial}{\partial \beta} \frac{\partial \beta}{\partial \phi_1} = \frac{1}{2} \frac{\partial}{\partial \alpha} + \frac{\partial}{\partial \beta}$$

$$\frac{\partial}{\partial \phi_2} = \frac{\partial}{\partial \alpha} \frac{\partial \alpha}{\partial \phi_2} + \frac{\partial}{\partial \beta} \frac{\partial \beta}{\partial \phi_2} = \frac{1}{2} \frac{\partial}{\partial \alpha} - \frac{\partial}{\partial \beta}$$

$$\frac{\partial^2}{\partial \phi_1^2} = \frac{1}{4} \frac{\partial^2}{\partial \alpha^2} + \frac{\partial^2}{\partial \beta^2} + \frac{\partial^2}{\partial \alpha \partial \beta}$$

$$\frac{\partial^2}{\partial \phi_2^2} = \frac{1}{4} \frac{\partial^2}{\partial \alpha^2} + \frac{\partial^2}{\partial \beta^2} - \frac{\partial^2}{\partial \alpha \partial \beta}$$

$$\hat{H} = - \frac{\hbar^2}{2mR^2} \left(\frac{\partial^2}{\partial \phi_1^2} + \frac{\partial^2}{\partial \phi_2^2} \right) =$$

$$= - \frac{\hbar^2}{2mR^2} \left(\frac{1}{2} \frac{\partial^2}{\partial \alpha^2} + 2 \frac{\partial^2}{\partial \beta^2} \right) =$$

$$= - \frac{\hbar^2}{2 \underbrace{(2m)}_M R^2} \frac{\partial^2}{\partial \alpha^2} - \frac{\hbar^2}{2 \underbrace{\left(\frac{m}{2}\right)}_\mu R^2} \frac{\partial^2}{\partial \beta^2} ,$$

$M = 2m$: mass of the center of mass,
with position $\alpha \in [0, 2\pi]$

$\mu = \frac{m}{2}$: mass of the reduced system,
with position $\beta \in [0, 2\pi]$

$$H' = H + V(\beta) =$$

$$= -\frac{\hbar^2}{2MR^2} \frac{\partial^2}{\partial \alpha^2} - \frac{\hbar^2}{2\mu R^2} \frac{\partial^2}{\partial \beta^2} + V(\beta) ,$$

which is separable in α, β .

Thus : $\Psi_{kl}(\alpha, \beta) = \psi(\alpha) f(\beta) ,$

where $\psi(\alpha) = \psi(\alpha + 2\pi) , f(\beta) = f(\beta + 2\pi)$

UNPERTURBED

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$$c) \quad \Psi_{kl}^{(0)}(\alpha, \beta) = \psi_k^{(0)}(\alpha) \psi_l^{(0)}(\beta) ,$$

$$\psi_k^{(0)}(x) = \frac{1}{\sqrt{2\pi}} e^{\pm i k x}$$

$$E_{kl}^{(1)} = \langle \Psi_{kl}^{(0)} | \hat{V}(\beta) | \Psi_{kl}^{(0)} \rangle =$$

$$\begin{aligned}
 &= \langle \psi_k^{(0)}(\alpha) | \psi_k^{(0)}(\alpha) \rangle \langle \psi_l^{(0)}(\beta) | \hat{V}(\beta) | \psi_l^{(0)}(\beta) \rangle \\
 &= 1 \cdot \int_0^{2\pi} d\beta \frac{V_0}{2\pi} e^{-il\beta} \cos \beta e^{il\beta} = 0
 \end{aligned}$$

(But) l and $-l$ give the same energy, so E_l is 2-fold degenerate $\rightarrow \psi_l, \psi_{-l} \rightarrow E_l$

$$\langle \psi_l^{(0)} | \hat{V} | \psi_l^{(0)} \rangle = 0 \quad (\text{see above})$$

AND

$$\begin{aligned}
 &\langle \psi_l^{(0)} | \hat{V} | \psi_{-l}^{(0)} \rangle = \\
 &= \frac{V_0}{2\pi} \int_0^{2\pi} d\beta e^{-2il\beta} \cos(\beta) \equiv 0
 \end{aligned}$$

So, the first order correction to the energy of any state is zero.

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